Routing Protocol or Route Type	Default External Preference
Direct	0
OSPF	10
IS-IS	15
Static	60
RIP	100
OSPF ASE	150
OSPF NSSA	150
IBGP	255
EBGP	255

NOTE

In Table 2-1, the value 0 indicates direct routes and the value 255 indicates routes learned from unreliable sources. A smaller value indicates a higher preference.

You can manually configure the preference of a routing protocol except direct routes. In addition, the preference for each static route varies.

https://support.huawei.com/enterprise/en/doc/EDOC1000047418/e3a3303d/routing-protocol-preference

Default Routes

Default routes are special routes used only when packets to be forwarded do not match any routing entry in a routing table. If the destination address of a packet does not match any entry in the routing table, the packet is sent through a default route. If no default route exists and the destination address of the packet does not match any entry in the routing table, the packet is discarded. An Internet Control Message Protocol (ICMP) packet is then sent, informing the originating host that the destination host or network is unreachable.

In a routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0). You can run the **display ip routing-table** command to check whether a default route is configured. Generally, administrators can manually configure default static routes. Default routes can also be generated through dynamic routing protocols such as OSPF and IS-IS.

https://support.huawei.com/enterprise/en/doc/EDOC1100086956/e18497a3/ip-routing-basics

Routing Table: Public
Destinations : 8 Routes : 8

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	D	10.1.4.2	GigabitEthernet1/0/0
10.1.4.0/30	OSPF	10	0	D	10.1.4.1	GigabitEthernet1/0/0
10.1.4.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.1.4.2/32	OSPF	10	0	D	10.1.4.2	GigabitEthernet1/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The routing table contains the following key information:

- Destination: indicates the destination address or destination network segment of the route.
- Mask: indicates the network mask. It is used together with the destination address (through an AND operation) to identify the address of the network segment where a destination host or router resides. For example, if the destination address is 1.1.1.1 and the mask is 255.255.255.0, the address of the network segment where the host or router resides is 1.1.1.0.
- Proto: indicates the method by which the route is obtained. For example, Direct indicates a direct route, and Static indicates a static route.
- Pre: indicates the preference of the route in the IP routing table. Multiple routes to the same destination may exist and have different next hops or outbound interfaces. These routes may be discovered by different routing protocols or manually configured static routes. Among these routes, the one with the highest priority (smallest preference value) is selected as the optimal route.
- Cost: indicates the cost of the route. When multiple routes to the same destination have the same preference, the route with the smallest cost is selected as the optimal route. The route cost is a value used to evaluate the quality of a route path. It is also called the route metric and indicates the cost of the route path from the source node to the destination node. The cost depends on multiple factors such as the network topology, bandwidth, delay, reliability, and security.
- Flags: indicates the source and destination of a data packet and how the router processes the packet.
- Nexthop: indicates the IP address of the next hop.

For the preference, same destination = same IP address + same mask

What Are the Rules for Selecting IP Routes?

Route Preference

Routing protocols (including static routing) may discover different routes to the same destination, but not all the routes are optimal. In this case, the one discovered by the routing protocol with the highest priority (smallest preference value) is selected as the optimal route.

The following table lists the default preference of each routing protocol or route type on Huawei routers. Value 0 indicates a direct route, and value 255 indicates any route learned from an unreliable source. A smaller preference value indicates a higher priority. Except for direct routes, the preferences of all routing protocols or route types can be manually configured. In addition, the preferences of static routes can vary.

Table 1 Routing protocols and their default preferences

Routing Protocol or Route Type	Preference
Direct	0
OSPF	10
IS-IS	15
Static	60
RIP	100
BGP	255

On Huawei products, such as the NE40E, both external and internal preferences are defined. An external preference refers to the preference manually configured for a routing protocol. The preceding table lists the default external preferences of routing protocols. If different routing protocols are configured with the same external preference, the system selects the optimal route by internal preference. The following table lists the internal preferences of different routing protocols.

Table 2 Internal preferences of different routing protocols

Routing Protocol or Route Type	Preference
Direct	0
OSPF inter-area	10
IS-IS Level-1	15
IS-IS Level-2	18
EBGP	20
Static	60
RIP	100
OSPF ASE	150
OSPF NSSA	150
IBGP	200

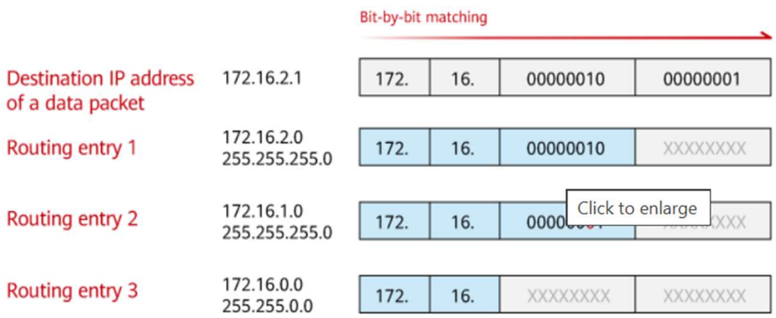
For example, if an OSPF route and a static route are both destined for 10.1.1.0/24 and their external preferences are both set to 5, the system selects the optimal route by internal preference. The internal preference of OSPF is 10, whereas the internal preference of the static route is 60. As such, the OSPF route takes precedence over the static route in route selection.

For the preference, same destination = same IP address + same mask. If there routes with same destination but different preference values, only route with the smallest preference will be installed in the routing table.

Longest match rule

The prefix of each IP route is defined by an IP address and mask length. When receiving an IP packet, a router matches the destination IP address of the packet against all entries in its local routing table bit by bit until the entry with the longest prefix match is found. The router then forwards the packet based on the next hop in this entry. The longest match rule is the most fundamental part of route selection. It improves the route selection efficiency and prevents routing loops.

Assume that there are three routing entries 172.16.2.0/24, 172.16.1.0/24, and 172.16.0.0/16 (1, 2, and 3 for short) in the routing table, and the corresponding outbound interfaces are S1, S2, and S3, respectively. To forward a data packet destined for 172.16.2.1, the router selects routing entry 1 according to the longest match rule because its prefix is longer and more accurate than the prefix of routing entry 3 and routing entry 2 does not match the destination address.



Default Route

Default routes are special routes. If the destination address of a packet does not match any destination address in the routing table, the router forwards the packet using a default route. If no default route is available and the destination address of the packet does not exist in the routing table, the router discards the packet. Default routes can be manually configured or generated by dynamic routing protocols, such as OSPF and IS-IS. In the routing table, a default route is presented with the destination address 0.0.0.0/0. You can run the **display ip routing-table** command to check whether a default route is available.

The IP route selection process is as follows:

1. After receiving a data packet, a router searches for a route that best matches the destination address of the packet according to a route selection rule (such as the longest match rule). If such a route is found, the router forwards the packet based on the outbound interface and next hop of the route.
2. If no matching route is found, the device checks whether a default route exists. If no default route exists, the device discards the packet and sends an ICMP message to the source device.

https://info.support.huawei.com/info-finder/encyclopedia/en/IP+routing.html

5.3.2 *Longest Prefix Match*

The longest prefix match (LPM) is a routing lookup mechanism used by default by almost all routers in the industry today.

Each entry in the routing table of a router specifies a network, and if these network addresses overlap, a destination address may match with multiple entries. When a router receives an IP packet, it compares the packet's destination IP address with all the routing entries in the local routing table bit by bit until it finds the entry with the longest match. This is the longest prefix match mechanism. It is called this because it is also the entry where the largest number of leading address bits of the destination address match those in the table entry.

Installation of routes in routing table